Sre Cibi Varsshan K

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Professional Summary

Graphics-focused programmer specializing in C++ and GPU-accelerated systems. Experienced in compute shaders, real-time rendering pipelines, and high-performance parallel computation. Strong foundation in algorithms, computer vision fundamentals, and systems programming. Capable of reading and implementing research-driven solutions across image processing, machine learning, and spatial computing domains.

Core Technical Skills

Languages: C++, C, C#, Python, TypeScript, Bash

Graphics & GPU Programming: OpenGL, Compute Shaders, Ray Tracing, Ray Marching, Real-Time Rendering Pipelines

Computer Vision & ML: Image Processing, Feature Extraction (ORB), Visual SLAM, Classification Models

Algorithms & Systems: Data Structures & Algorithms, Parallel Computing, Memory Management

Tools & Platforms: Linux, Unity, Flask

Selected Projects

GPU Aided Bulk Image Encryption (ChaCha20) *C++, OpenGL*

- Designed and implemented a high-throughput image encryption system using ChaCha20 stream cipher on GPU via OpenGL compute shaders.
- Developed unified buffer architecture using Shader Storage Buffer Objects (SSBOs) for batch processing of large image datasets.
- Achieved parallel encryption of gigabyte-scale data using ARX-based keystream generation mapped to GPU threads.
- Engineered metadata-driven reconstruction pipeline enabling lossless decryption of images.
- Demonstrates applied cryptography, parallel algorithms, and GPU compute optimization.

Ray Marching Renderer (Compute Shader Pipeline and fragment shader pipeline) *C++, OpenGL*

- Built a GPU-driven ray marching engine using compute and fragment shaders for procedural 3D rendering.
- Implemented signed distance functions, lighting models, and real-time shading pipelines.
- Designed for minimal CPU overhead with fully parallel GPU execution.

Ray Tracer (Physically Based Rendering) *C++*

- Developed a CPU-based ray tracer implementing core rendering principles including reflections, materials, and global illumination.
- Focused on physically based rendering and light transport simulation.

ESP32 ORB-SLAM Pipeline *Python, C/C++*

- Developed a visual SLAM system streaming real-time data from ESP32 camera hardware.

- Implemented ORB feature detection, tracking, and spatial mapping.
- Explored integration between embedded systems and computer vision pipelines.

UAV Central Dashboard *TypeScript*

- Engineered a real-time multi-UAV monitoring and coordination dashboard.
- Implemented telemetry visualization and spatial tracking systems.
- **Patent (Applied):** Autonomous multi-UAV system for bandwidth-constrained environments using deterministic state synchronization, delta state serialization, and client-side architecture.

DSA_H (C Library) *C*

- Developed low-level implementations of data structures and algorithms with emphasis on memory efficiency and pointer manipulation.

Additional Projects

Breast Cancer Classification Model *Python*

- Built supervised ML models achieving 95% accuracy using Random Forest, SVM, and Logistic Regression.
- Applied preprocessing, feature selection, and evaluation metrics.

AI Image Generation Web App *Python, Flask*

- Developed a web-based system for real-time image generation using diffusion models.
- Implemented backend pipeline for prompt handling and API-based inference.

Deep Q-Network Space Shooter (Reinforcement Learning) *Python, PyTorch*

- Implemented a reinforcement learning agent using Deep Q-Networks (DQN) to autonomously play a 2D space shooter environment.
- Designed and trained a convolutional neural network to learn action-value functions directly from visual state inputs.
- Applied core RL techniques including experience replay and target network stabilization for improved training convergence.
- Integrated environment simulation and reward shaping to optimize agent performance and survival strategy.
- Demonstrates applied machine learning, algorithm design, and decision-making under dynamic environments.

Education

VIT, Vellore, India
B.Tech in Computer Science (2027)

Relevant Coursework: Computer Graphics, Data Structures & Algorithms, Machine Learning, Computer Vision